

Chapter 6 — Advanced topics and future directions

Notebook EDA on static CSVs is the baseline; large and streaming datasets push new tooling and pipeline integration

Learning objectives

- Name trends in automated EDA, outline challenges for streaming exploration
- Describe how EDA hooks into ML pipelines

Automated EDA tools

- Report generators and AutoML front-ends accelerate first passes on wide tables
- Fairness implications connected to Chapter 7

EDA in big data and streaming

- Batch assumptions fail when data never stops arriving
- Sliding windows, online summaries, and sampled visualizations replace full-table scans
- Latency and consistency constraints mirror Chapter 5 streaming preprocessing themes

Integration with ML pipelines

- Exploration informs feature stores, train-validation splits, and monitoring baselines
- Documented EDA artifacts become regression tests when production data drifts

Looking ahead

- Chapter 7 deepens bias and fairness analysis once exploration reveals segment disparities
- Strong EDA logs make later modeling and auditing more trustworthy

Takeaways

- Automation scales the first pass; humans keep domain judgment
- Streaming and pipeline integration extend EDA beyond one-off notebooks

Next

- Complete the quiz for this part
- Pause for the quiz